

Hero Companion

Help & Release Notes — your City of Heroes sidekick

App version 0.12.18 · Model v30 · Game database 2026.1.1242

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Hero Companion is a fan-made build planner for City of Heroes: Homecoming. It plans, optimizes, and walks you through character builds using the game's actual math — to-hit, resistance, defense, endurance, recharge — not templates or guesswork. It runs entirely on your own computer, next to (never inside) the game.

Quick Start

Open the app and pick one of the entry cards:

- **Build a new level 50** — you choose archetype, powersets, the content you'll run, and your role on a team; the optimizer designs a complete end-game build with enhancement slotting.
- **Start from scratch (level 1)** — same choices, but you get a level-by-level leveling companion that walks every pick and slot from 1 to 50.
- **Import from Mids Reborn** — paste a Mids build (or load a .mbd file) and the tool evaluates it, then can optimize from there.
- **Import a character you play** — type `/build_save_file`` in game, then click "Find my characters for me": the tool locates your Homecoming saves itself and imports the one you pick. (Unusual install location? Tell it your game folder once and it remembers.)

Everything is reversible. Nothing you click is permanent, and in-game a `/respec` rewrites any character anyway.

The Build Panel

Pick your archetype, primary, and secondary. The panel fills with your powers; click a power to add or remove it, and use the slot controls to place enhancement slots (the game grants 67 placeable slots across a career; the tool enforces the real limits).

The stat panel updates live: defense by position, resistance by type, recharge, recovery, damage. Caps are shown — defense soft cap 45%, resistance cap by archetype.

Content and Role

These two dropdowns are the most important choice in the tool:

- **Content** — what you actually run: general play, task forces, incarnate trials, fire farms, PvP. It sets what the build must survive and deliver.
- **Role** — what you're there to do: damage, tank, buff/debuff support, control,

healing. The optimizer maximizes your role output, not a generic score.

A support character is scored on how much its debuffs and buffs actually change fights (magnitude times uptime), not on its own survival. That's deliberate: the tool builds characters that are noticed as contributors.

The Optimizer

The "Optimize" actions run a real search over powers, pools, and slotting:

- It explores every legal add, drop, and swap — including all power pools — and runs until no single change improves the build (with honest certificates saying so).
- It respects the game's rules: pool limits (four max), the origin-pool rule (only one of Sorcery / Experimentation / Force of Will / Gadgetry / Utility Belt), archetype set access, prerequisite tiers, and level availability.
- It learns: strong past results seed future searches as champions.

If a suggestion looks odd, it earned its place in the math — but you always have the final say. Swap anything; the tool re-evaluates around your choices.

Leveling from 1 to 50

For from-scratch characters, the leveling walk shows exactly what happens at every level on the real Homecoming schedule: a power on even levels, slots on odd levels, pools at 4, epics at 35, respec trials at 24/34/44, and cost-smart enhancement advice for each stretch (cheap early, common IOs from 7, sets near the end).

You can take every suggestion or none of them. The walk tracks where you've deviated and offers to re-fit the end-game around your actual choices.

Kheldians (Peacebringer / Warshade)

Kheldians follow their own rules and the tool knows it: inherent flight from level 1 (no travel pool needed unless you want one), Nova and Dwarf forms inside your own power sets, and no epic pool at all — the walk and the wizard both reflect that.

Arachnos Soldiers and Widows

VEATs live a two-phase career and the walk follows it honestly:

- **Levels 1–23:** only your base sets (Arachnos Soldier / Training and Gadgets, or Widow Training / Teamwork) — branch powers can't be taken yet, so the walk never suggests them.
- **Level 24:** the mandatory respec. You choose your branch — Crab or Bane, Night Widow or Fortunata — and re-place every pick from level 1 with all six sets open. The walk hands you the complete re-place order right at this step.
- **Levels 24–50:** the walk continues from the respec order, branch powers included.

Enhancement Converter and Haul Appraiser

The Converter panel answers two questions:

- **"How do I get this IO cheaply?"** — a concrete cheapest path per enhancement: which piece to buy or craft, which converters to use, and the converter/merit cost. One rule of the game worth knowing: conversions can't jump from cheap pools into purples — the tool's paths never pretend otherwise.
- **"Is this drop worth anything?"** — paste your drops (recipes included, straight from the game). Each item gets a verdict: keep for your build, craft-then-convert for profit, or just sell.

Saving, Importing, Exporting

- **Save** keeps the character's plan and leveling progress locally so you can resume. Auto-save runs in the background.
- **Import/Export** is Mids Reborn compatible — bring builds in, take builds out. Your builds are yours; nothing leaves your machine.

Hero or Villain

The alignment button in the banner reskins the whole app — Hero Companion in blue, Villain Companion in red. Pure style; your choice is remembered.

Bugs, Champions, and Updates

Hero Companion is a living tool, and you're part of how it improves. Three buttons, all strictly opt-in — the app never sends anything on its own:

- **Report a bug** — opens a pre-filled bug report on the project's GitHub page with your app, model, and game-data versions already included (that's usually the hard part of a good bug report). Add what happened and post it. Nothing is sent until you click submit on GitHub itself.
- **Submit champion** — think your current build beats the shipped champion for its archetype and role? This saves your build as a *champion candidate* file and opens the submission queue. The development hub re-scores every candidate with its own math — if your build genuinely wins, it becomes the new shipped champion in a future update, with credit to you.
- **check for updates** (bottom of the page) — compares your version against the latest release on GitHub. On first run the app asks once whether to do this check automatically at startup (it contacts github.com to compare version numbers and sends nothing else); say yes and new releases greet you with an "Update now / Remind me later" banner. Say no and the footer button is the only check that ever runs. Updates are never downloaded or installed for you.

A few words the tool uses precisely: a **champion** is the best certified build for an archetype + powersets + role combination — champions ship with the app and give the optimizer its head start. A **candidate** is your claim to beat one. The ****model version**** (v23 today) stamps which generation of the scoring physics produced a number, and the **data version** stamps which game database it was computed against — so

scores are always compared apples to apples.

Frequently Asked

The tool and the game disagree — who's right? The game, always. The tool's model is verified against the Homecoming Wiki and City of Data, but patches happen. When you find a disagreement, that's a bug worth reporting.

****Why won't it give me a second origin pool / a branch power at level 10 / an epic on my Peacebringer?**** Because the game won't allow it. The tool only suggests choices that exist for your character at that level.

Does it play the game for me? No. It never touches the game client or servers. It's a planner you read while you play.

Is it free? Free and noncommercial, forever (CC BY-NC-SA 4.0). See the Terms of Use, License, and Credits links at the bottom of the app.

Credits

Built on the work of the City of Heroes community: NCSoft (the game, all rights theirs), Homecoming Servers, Mids Reborn (all game data, parsed from your own install), the Unofficial Homecoming Wiki, City of Data v2, the Homecoming Forums, the Paragon Wiki Archive — Guyver [SoV] of the Sovereign supergroup, whose openly shared master builds were the standard this tool's optimizer had to honestly beat — and Maelwys, whose expert forum reviews caught what our own tests missed, round after round.

Hero Companion — What's New

0.12.18 — 2026-07-10

- **Your core armor toggles are enhanced like they deserve (model v30).** The

optimizer used to treat a met survival target as "done" — past that point, enhancing your strongest shields was worth literally nothing to it, so powers like Fire Shield or Temp Invulnerability could end up as two-slot parking spots for global IOs while a free minor power got a full set. That threshold is gone: survival keeps its real, continuously-measured value all the way to your archetype's caps, so the strongest armor powers get their resistance and defense aspects enhanced first — with endurance reduction riding along — and the weaker or free powers take the mule duty. Global IOs now also prefer the cheapest hosts, so an expensive toggle like Weave keeps its slots for real enhancement. (Thanks to Maelwys for the precise field report that pinned this down.)

- ****Ten set-bonus families the planner literally could not see now count**

(model v30):** knockback protection, slow resistance, all six mez-duration bonuses (confuse, hold, stun, sleep, immobilize, fear), movement speed, range, and endurance discount — 103 bonus tiers back-filled straight from the game client's own data, verified value-for-value. They appear in your totals and on every enhancement card (the "not yet in totals" note is gone), and the optimizer's scoring now values knockback protection, slow resistance, and your controls' longer mez durations. The three stackable knockback IOs (Karma, Steadfast Protection, Blessing of the Zephyr) are priced too — mag 4 each, exactly as the game grants them.

- **Preview toggles now stack — every combination the game allows.** The

one-preview-at-a-time rule blocked real play: Build Up + Aim is the classic opener, Farsight + Group Invisibility + Power Boost is how a Time build checks its soft cap, Meltdown + Shadow Meld is a genuine defensive layer. All of it now stacks freely. Honesty replaced restriction: short strike windows (Build Up class) wear their own chip, and while any are checked the totals panel says loudly that you're looking at a BURST VIEW — a window of seconds, not what the build sustains. Cycling buffs (Hasten, Farsight class) keep the chip with their real uptime at your build's recharge — and a short-window power your recharge makes effectively permanent (Parry spam) earns the cycle label its math deserves. Also fixed: unchecking behavior could leave a chip LOOKING checked while its power no longer counted — chips now always repaint with their true state.

0.12.17 — 2026-07-10

- **The "count in totals" checkbox now matches what the game actually allows.**

It used to be one uniform Σ checkbox on every power — which lied for most of them. Now there are three honest states, each with its own color and glyph (never color alone): always-on powers (autos, passives, invariants) show a locked badge with no control at all, since the game gives them no off-switch; toggles keep a checkbox (, green, default ON) as their real home — uncheck a mule host or run a quick "what if Weave were off"; timed self-buff clicks (Hasten, Build Up, and their kind) get a preview toggle (, orange, default OFF) that previews one window at a time, since the game only ever has one of these running as a meaningful "on" state — checking one switches any other preview off. Plain

attacks get no chip at all. Also: the endurance-sustainability warning now respects your checkbox choices (an unchecked toggle no longer counts against your endurance), and it actually shows up in the totals panel when your checked toggles + attack chain would run you dry.

- **Customize the targets the solver chases.** "Customize build targets..."

(on the play-style summary and next to the Content/Role pickers) opens an editor seeded from your chosen preset: typed AND positional defense, per-type resistance, recharge, recovery, regeneration, and max HP. Set a number to chase it, clear it to drop it — a resistance-primary set like Fiery Aura can finally chase 90% S/L/F resistance plus recharge and HP instead of being held to defense-shaped preset thresholds. Values clamp to game reality (your archetype's own resistance cap); when the ask exceeds what 67 slots can buy, the solve gets as close as it can and says so. Custom targets persist with the save, and you can keep them as named presets of your own to reuse on any build. Honest labeling throughout: a custom-target build is YOURS — it never reads as a certified champion build. Also fixed on the way: resuming a saved character now restores its Content/Role picks into the dropdowns (they used to come back empty, so a re-solve could silently run against different targets).

- **The same set piece can no longer be slotted twice in one power.** The

game doesn't allow it, and now neither does the picker: pieces already in the power show "slotted here" and can't be picked again. If a build arrives with duplicates anyway, validation flags it as a real error, the duplicate no longer conjures set-bonus tiers (tiers count distinct pieces, so the card's "N of 6" and your totals always agree), and stackable globals like Luck of the Gambler +recharge now respect the game's rule of five — a sixth copy grants nothing, exactly like in-game. Identical Hamidon/Titan/Hydra Origins and D-Syncs still stack freely (the game's actual rule).

- **Preview Enhancement Boosters before you buy them.** Every boostable

enhancement's detail card now carries a booster stepper: walk it from +1 to +5 one level at a time and watch your real totals move — "+3 reaches the soft cap, +4 and +5 buy nothing here" is now something you can see instead of guess. Previewed pieces show their level exactly like imports do ("50+3") but in a distinct preview color, and the totals panel says plainly that previews are not saved as owned. Pieces that can't boost say why: attuned enhancements (including ATOs and Winter sets) scale with your level instead and keep exemplaring down, and Hamidon/Titan/Hydra Origins and D-Syncs get their strength from their own level. The card also explains where boosters come from and the exemplar trade-off in plain English.

- **Power Boost previews for real.** Powers that amplify your other buffs

(Power Boost, Power Build Up) now have their own preview chip (): switch one on and your buffable defense and ToHit values amplify exactly the way the game does it — effects the game marks "Ignores Buffs" are skipped, verified per-effect against the game's own data. Combine it with one buff preview (Power Boost + Farsight) to answer the real question: does this build actually hit the soft cap in its burst window? The totals panel labels the result as a burst view, never sustained. (Clarion Radial isn't modeled yet — its incarnate data doesn't carry a verifiable amplifier record; the tool says so rather than guessing.)

- **Running toggles are no longer called "mules."** A toggle or always-on

power that hosts global uniques (Fire Shield with a Steadfast, Weave with a Luck of the Gambler) now reads "Global host — this toggle runs; its slots carry build-wide bonuses" instead of "Global mules,"

which implied the power was dead weight. True set-real-estate powers keep honest mule wording.

- **The detail card grew into a real rail.** The right-side panel now scrolls

on its own (a long six-piece card browses in place while your build stays on screen), every slotted enhancement carries its own — hover the chit — so the full detail card is one click away without entering the change-picker, and both rail views are LIVE: slot a new piece of the displayed set and the roster chips and lit bonus tiers update in place, no reopening. The power info panel now shows the full merchant-style set view too: one set in a power renders its card inline; a mixed-set power gets one compact row per set ("Shield Breaker 1/6 — next: +2.5% recovery") that expands to the full card. Set-bonus tiers also read cleaner — repeated per-type values collapse to one line, and bonuses the game grants but the totals can't count yet (slow resist, movement, range, mez duration) say so honestly instead of showing a bare name.

- **The true in-game enhancement experience.** Click a slotted enhancement and

open its full detail card: the game's own description text with honest numbers at YOUR piece's level and boost state ("Enhances the defense debuff potential of a power by 17.4%..."), the attuned and unique rules, and the complete parent set — every piece marked slotted-here / elsewhere / missing, and every set-bonus tier lit as you attain it, with "next piece" showing exactly what one more piece buys. All text extracted from the game client itself (227 sets, every piece and tier), not paraphrased. The picker now shows what's currently slotted with a "Full details" button, and power cards reveal a hint so the info panels are discoverable.

0.12.16 — 2026-07-09

- ****Every "How do you play?" question explains itself the moment you answer**

it.** The Role and "You fight from" pop-ups were silent on the first pass through the wizard (they come before "Mostly in", and the explainers waited for that answer). Now each question pops its tailored explanation as soon as YOU answer it, on every surface that asks — Start New, Respec 50, and Change how you started — while the combined summary still waits until your Role and Content are actually chosen. Also: the import screen now says plainly that in-game /build_save_file exports carry no boost levels — import a Mids .mbd to bring +1..+5 and HO levels across.

- **Henchmen inherit your set bonuses — and the optimizer knows.** The game

gives Mastermind henchmen half of your true set bonuses (verified straight from the game client): +Max HP arrives as a flat amount from the Mastermind's own hit points, identical for every henchman, so your squishy tier-1s gain the most staying power — exactly the effect the forum discussion described. Builds are now scored with each henchman's real survival in the fight (tier hit points from the game's own tables, deaths cost the actual resummon and re-upgrade cast times), so +HP, defense and resistance bonuses carry honest extra value on a Mastermind. Globals like Unbreakable Guard and accolade HP do NOT reach henchmen (also the game's rule), and the planner never credits them there.

- **Heal-strength set bonuses exist now.** The game's 11 heal-strength bonuses

(Numina's 4-piece +6% heal, Doctored Wounds +4%, Panacea's 6-piece +6%...) had parsed to nothing — same story as the accuracy bonuses fixed in 0.12.15. All 11 are restored from the game client, the totals panel shows your +% Heal strength, and the optimizer values what they do to your actual healing

output.

- **Slot anything by hand.** The enhancement picker now offers the single enhancements a power accepts alongside its sets: common crafted IOs, Hamidon / Titan / Hydra Origins, and D-Syncs. Identical copies stack freely (that's the game's rule for these), totals price them exactly, and they round-trip through Mids export/import.
- **In-combat view (suppression).** A new Stats toggle shows your totals as they are mid-fight: powers like Stealth lose their suppressible defense the moment you attack or get hit, exactly per each effect's own suppression flags — same as Mids' Options > Effects and Maths > Suppression. Display only: builds are optimized on the same numbers either way.
- **Real enhancement levels from imports.** Boosted IOs (50+5) and level-53 Hamidon Origins now import with their true levels, show as "50+5" / "53" on the slot, and count in the totals at the game's real math (+5% per boost level; a 53 HO is worth 38.3%, not 33.3%). Fixed a one-off: Mids stores levels 0-based, so every imported IO used to read one level low. Exports to Mids now carry boosters back too.
- **Cards say where the res job went.** A base-slotted power that carries its own resistance debuff (Melt Armor and friends) now names the power actually holding the res procs and why it won the job, instead of a vague "budget went elsewhere."
- **No more duplicate epic pools.** Three pools leaked into the wrong archetypes' dropdowns by stale upstream data (Stalkers and Dominators each saw two same-named "Mastery" pools). The game client's own eligibility rules say who gets what; the extra entries are gone.

0.12.15 — 2026-07-08

- **Fresh gold-standard champions, certified on verified game data.** All five champion builds were re-converged from scratch under this release's corrected model and data — every value they stand on now checks against the game client itself. Old champion scores looked bigger; these are the honest numbers.
- **The wizard never chooses for you.** All four "How do you play?" questions start unanswered and the Build button waits until you've answered them — the planner never invents your playstyle. Each answer is tagged with where it came from ("your pick" / "from your setup"), and reopening the wizard restores what you chose — including your travel power. Travel is never auto-picked: endgame content entry can require specific travel (BAF and Lambda admit only Flight or Teleport), so that call is always yours.
- **New Role option: Mixed role / Generalist.** The honest choice when you don't specialize — balanced targets for your content, judged by overall contribution instead of one role's lens. Its pop-up explains the trade plainly.
- **Slotting notes are easier to find and use.** The whole note chip on a power card is now the button (hover for the bonus summary, click for the reasoning), with a one-time hint on your first build. Power-card tooltips lead with the full power name, so truncated names are readable. The / + slot buttons explain the shared 67-slot pool, and the slot counter tells you when you have free slots to spend. The power-info panel closes on Esc or a click outside.

- **Global Accuracy set bonuses now exist — and are valued.** A parser gap had dropped every global-accuracy set bonus in the game (Luck of the Gambler's 4-piece +9%, Adjusted Targeting, the archetype sets' +15%...) — 65 bonuses, invisible to the totals and the optimizer alike. All 65 are restored straight from the game client's own data, the totals panel shows your +% Accuracy, and the optimizer now prices accuracy by the game's to-hit math: against +3/+4 enemies (iTrials, hard content) accuracy multiplies your real damage until you reach the 95% hit ceiling, so builds for that content chase it — and builds for even-level content don't waste slots on it. The data reality-check now verifies 282 set-bonus values against the live game (a name-matching bug had silently limited it to 43), all matching.
- **The "How do you play?" questions now explain themselves.** Every choice in the build wizard pops a detailed explanation written for YOUR character — a Mastermind's "Damage dealer" talks about henchmen and pet sets, a Super Reflexes scrapper's iTrial preset explains why it chases positional instead of typed defense. A new summary panel shows exactly what your combined answers make the planner chase (the real defense, resistance, and recharge targets), so nothing about the build's direction is a mystery before you hit Build.
- **Credits: Maelwys.** His expert reviews on the Homecoming forums caught what our own tests missed, round after round. Added to the in-app credits and CREDITS.md.
- **Common IO icons are back.** 0.12.13 accidentally dropped every common IO's icon, so a generated build's Accuracy/Damage/Endurance IOs rendered as empty-looking slots next to expensive globals. The same defect gave Hamidon Origins a level they don't have, which scaled imported low-level HOs toward zero. Both fixed, and the audit now fails hard if any slotted piece ever ships icon-less again.
- **Hamidon Origins stack legally.** The validator warned "duplicate piece slotted 2x" on double-Nucleolus cores. HOs, Titan/Hydra Origins, and D-Syncs are not set pieces — the game lets you slot as many identical copies as you like, and the planner now agrees.
- **Real attacks reach 6 slots.** The optimizer always wanted six, but a budget accounting gap plus an order-blind trim kept every real attack at exactly five while filler powers kept six. Attacks now get their sixth slot; any trimming falls on junk fills and filler powers first, never the rotation.
- **Proc bombs carry accuracy.** Big proc bombs reserve a slot for a Nucleolus Exposure (Accuracy/Damage, recharge-free so every proc keeps its full chance) — and bombs no longer shrink a 6-slot power down to the proc count.
- **No more orphaned set pieces.** The res anchor, Force Feedback seating, and the endurance-relief pass all refuse swaps that would strand a lone set piece with no bonus, eat a global, or overwrite a proc. Force Feedback now lands in a spammed knockback attack's spare slot instead of cannibalizing a proc bomb.
- **Signature support buffs always work.** A damage-role Mastermind's Barrier Reef shipped as a bare 1-slot mule; signature buff clicks from a support set now always carry a working set, whatever role the build solves for.
- **Max HP totals are honest.** A /Regen build displayed "+58913% Max HP" (flat hit points summed as a percent). Totals now show the real number — "+79.9% = 2410 HP (capped; 3286 uncapped)" — plus regeneration in HP/sec and recovery in end/sec, the way the game reports

them.

- **+HP set bonuses were 10x too big.** Verified against the game client's own set-bonus powers: a "Large Increased Health Bonus" is ~1.88% of base HP, not 18.75%. Every +HP set bonus was inflated tenfold in the totals and in the optimizer's HP math. Two Touch of the Nictus pieces now read +1.87%, a third Luck of the Gambler +1.12% — matching the live game exactly.

- **Card notes now describe the actual decision.** Slotting chips say "5 of 6 Red Fortune — earns +5% recharge, +1.5% fire resistance" instead of a vague "full set for its bonuses"; a couple of procs is labeled "Procs", not "Proc bomb"; and Hamidon-Origin proc hybrids get their own "Proc hybrid" note explaining the pattern (these cards previously showed nothing and could wrongly trigger the respec nag — that misfire is gone).

0.12.14 — July 7, 2026

- **Synced with the game's July 7 update (Issue 28, Page 3).** The patch normalized Dark Armor's Obscure Sustenance to a 60 second recharge for every archetype; our data carried the old 180 seconds for Brute, Scrapper, Stalker, and Tanker. A full diff of all 10,708 powers against the patched client confirmed these four values were the patch's only data change — archetype tables, pet classes, and proc rates all verified unchanged.

0.12.13 — July 7, 2026

- **Resistance procs for every role that can carry them.** Achilles' Heel, Annihilation, and Fury of the Gladiator multiply the whole spawn's incoming damage — and a damage role owns the biggest share of that damage. The optimizer now hunts a home for each res proc in every build (debuff toggles first, then the biggest eligible power), not just on debuffers and controllers. Annihilation's proc rate was also corrected against the game's own data.

- **Force Feedback: Chance for +Recharge finally counts.** The famous knockback proc is seated in your spammiest knockback attack and honestly valued: the average recharge it sustains (chance per cast × 5 seconds ÷ how often you actually fire) flows into everything recharge touches — attack chains, Hasten uptime, debuff cycling, pet resummons. The build totals show exactly how much of your recharge it carries.

- **Hamidon Origin enhancements modeled.** All 62 special enhancements are now priced correctly when slotted, and the optimizer uses the master pattern: a proc-hybrid attack's filler core trades up to two Nucleolus Exposures (about 66% accuracy and 66% damage in two slots, with no recharge to depress the proc rates). Premium set cores are left intact — those bonuses are build-defining.

- **Champion builds re-converged** under the corrected model and shipping in this update, plus a fix to pet resummon timing that slightly undercredited global recharge on timed summons.

0.12.12 — July 7, 2026

- **Henchmen and pets priced like the live game (the big Mastermind fix).** Pet damage was built on a stale snapshot that rated henchmen more than twice as hard-hitting as the real game, counted a single pet where the game spawns a squad, and treated every summon as permanent. All three are fixed from the game's own data: henchman damage now uses the live tables, Soldiers count as 2 Soldiers + 1 Medic (tier squads of 3/2/1 across all six Mastermind primaries), timed summons like Spiderlings earn only the share of a fight they're actually up (recharge slotting shortens the resummon wait), and powers whose pets don't inherit slotting no longer get phantom enhancement. The optimizer values pets on the squad's real, uptime-weighted output — the "healing mule" Mastermind builds die here.
- **Champion builds re-converged under the corrected model.** Every gold-standard champion was re-solved to full convergence with the accurate pet math and ships in this update.
- **Live play capture improvements** (shared with Companion Lite): the server pulse counts formations rather than repeated recruiting shouts, Positron Parts 1 and 2 are distinguished in every spelling, your characters' servers are detected automatically from the game's roster file, and the recruitment-channel list ships in the updatable content pack.
- **Your server shows on the board — detected automatically.** The game itself keeps a roster file (playerslot.txt) naming each character's server; Lite looks your characters up there the moment they appear in the log, so the live board's "Servers reporting" line fills in with zero configuration. Only the characters actually seen in your logs are recorded — the roster itself is never stored or uploaded.
- **Positron Parts 1 and 2 finally count separately.** Recruiting-alias matching is now longest-first, so "posi 2" no longer collapses into generic Positron — in every spelling recruiters use (posi2, pos 1, 1st posi, ...). The same fix protects every multi-word alias (ice mistral, mort kal, kmitf).
- **The recruitment-channel list ships in the content pack,** so shard global channels (where iTrials and raids organize) can be added as data. Private channels (tells, whispers) remain excluded unconditionally.

Companion Lite 0.1.13 — July 7, 2026

- **Companion Lite now feeds the live Pulse Boards directly.** No accounts, no tokens, no setup, no publish button: while it runs, captured play uploads over HTTPS to the project's locked storage and the board pipeline imports it into the live page within minutes. Uploads are incremental (only new play), tagged by an anonymous install id — account login names are replaced with meaningless codes before anything leaves the machine, and machine names or paths are never read at all. The only thing that ever becomes public is the rendered board itself.
- **The terms are the consent.** First start shows the terms (also under About): using Companion Lite means you accept that your captured play feeds the live board. If you don't accept, quit and uninstall. Turning game logging off (/logchat) stops capture any time.

- **A simpler tray.** Open the live Pulse Board, the in-game logging setup, Status, About (terms), Check for updates, Quit. The local board page and every publish or connect control are gone — Lite captures and feeds, the live board is the board.

Companion Lite 0.1.12 — July 7, 2026

- **Your Pulse Board now lives online, not on the game machine.** On the owner's machine (the one with a publish token), Companion Lite keeps the live web page current automatically — publishing at most every 15 minutes and only when new play was captured — and "Open Pulse Boards" opens the online page directly. Machines without a token are untouched: their board stays a private local page and Lite never talks to the network.
- **Published boards are scrubbed before they leave the machine.** The public variant drops the local file path and never shows account login names — an unnamed character appears as "Unnamed character #1" until capture learns its name. Character names (which are public in game) are kept.
- **The server pulse counts formations, not shouts.** A recruiter repeating "lf2m dfb" until the team fills is one formation, not a sighting per shout: repeated asks for the same content collapse into one, the recruiter saying "full" closes it (their next ask is a new run), and 10 minutes of silence does too. Dual-boxing is covered by the same rule — two of your clients logging the same shout count it once.
- **Money stays off the public board.** Influence earned/spent and market flow show only on your private local board — how much anyone makes is not community data. The future price board will use per-item prices, never personal wealth.
- **Levels, badges, and merits wait for the character sync.** A fresh capture only sees a sliver of a multi-year character, so achievement state stays off the public board until a one-time in-game character sync (level, badges, accolades, vet levels) can bring a character over whole. Public character cards show play since capture began: days seen, XP, drops, defeats.
- **"Recent formations" actually shows recent ones.** The card was frozen on the first 20 sightings ever captured; it now rolls with the latest 20.

0.12.11 — July 6, 2026

- **Sharper end-game builds from a fully re-converged optimizer.** The best-known build for each context was re-solved to true convergence under the current model and the corrected proc data, so the head start the tool builds from is stronger.
- **Live play capture (opt-in, private messages excluded).** With chat logging on, the app turns your sessions into insight — your own rewards and drops, plus what's forming on your shard from the public recruitment channels. It learns your server's shorthand as it reads. Private messages (tells and whispers) are never captured, and nothing leaves your machine.

- **Long-recharge attacks and holds now get the proc treatment (model v25).** A

single-target power like Seismic Smash or Dominate keeps a 2-3 piece accuracy/damage core and fills its remaining slots with damage procs when the proc math favors it — the classic expert pattern. The kept core avoids recharge pieces on purpose (slotted recharge lowers proc rates). Pet summons are never raided; their slots stay the pet's.

- **Proc rates corrected from the game itself.** Every damage proc's rate now comes straight from the current game data; 11 were wrong, including the archetype set procs, which were undervalued by up to 43%.

- **The respec suggestion now notices when the optimizer has learned.** Resuming a build made under an older version offers "Built under an older optimizer — see what's improved?" even when nothing is structurally wrong with the slotting. Build the plan to see whether a respec is worth it; if nothing meaningful improves, it says so. Saying "no" is remembered for that character until the optimizer actually changes again.

- **IO levels show on resumed builds again.** Resuming a saved character could show set pieces without the level badge under their icons (while common IOs still showed 50). Saves are now healed as they load, so every existing save gets its levels and icons back without re-saving.

0.12.10 — July 6, 2026

- **Pet summons are slotted for the pet now.** A summon power — Mastermind henchmen, a controller or Arachnos pet, Gang War, Phantom Army — was sometimes being used as a parking spot for the player's own always-on healing IOs (Numina, Miracle, Panacea) instead of a set that actually helps the pet. Those healing IOs belong in Health, where they buff you, so that's where they go now. Every pet summon is reserved for a real pet set — its accuracy, damage, and the pet enhancement bonuses (and the Mastermind archetype IOs) — and is forced to take one rather than sitting on filler. Thanks again to Maelwys for the example. Deeper slotting quality (procs versus full sets by attack, minus-resistance procs, when recharge in a summon actually matters) is still being tuned.

0.12.9 — July 6, 2026

- **Game data brought current.** Our numbers came from a Mids database that turned out to be about six months old, so recent balance patches were missing. We now reality-check against the live game and fixed everything that had drifted: the Brute, Sentinel, Tanker, and Dominator defense/resist/damage modifiers (this is why Brute defense was reading low), a batch of power endurance/recharge/range values, and slotting rules — including Mastermind archetype IOs now being slottable in personal attacks. Set bonuses were checked and are current. Thanks to Maelwys on the Homecoming forums for catching the stale data.

- **The slotting tags now show on any build, not just after Solve.** Resume or import a build and each power shows why it's slotted the way it is (full set / proc bomb / global mules) right away — previously those tags only appeared right after solving, which made a loaded build look like the update hadn't taken.

- **Respec worksheet.** When you resume a build with slots not earning set bonuses, a small bar appears at the top of the build — "Ready for respec?" — that opens the full plan as a pop-up (so it's there when you want it, out of the way when you don't). The plan shows every change power by power — the old slotting struck through, the new below it — the stat gains, and a grocery list split two ways: what to craft or buy, and what to unslot and sell (with a note on each item's worth). Every line has a checkbox to track your progress, and the whole thing is saved to the character, so you can work it over several sessions of crafting (the bar shows "N/M done"). Apply it to your build when ready, Undo to revert, or mark it completed. It never appears on a build that's already well slotted.
- **Explanations everywhere.** The slotting tags (proc bomb, full set, frankenslot, global mules) and the respec card now have a ? that opens a plain-language explanation of what the term means, why that approach is used, and your options — including why, for example, a frankenslot can beat a single six-piece set.
- **Updating no longer leaves a ghost tray icon, for real this time.** When the app updates itself, it now removes its own tray icon cleanly before the installer takes over, instead of being force-killed and leaving a dead icon behind that only clears when you mouse over it. (0.12.8 improved this for extra copies; this closes the main path — the app replacing itself.)

0.12.8 — July 6, 2026

- **Watch multiple accounts at once — per-character stat cards.** Dual-boxers can now tick more than one account, and the Play Log shows a card per character side by side (Rattle's kills and haul next to your farmer's), each with its own fit link. Accounts are toggle chips — click to add or remove one from what's watched.
- **The Play Log moved up into the main workspace.** Instead of a detached strip far below everything, it now docks in the wide build column — filling that space when no build is loaded, and sitting neatly under the build when one is. On narrow screens it still stacks full-width like the rest.
- **Cleaner haul advice.** Incarnate salvage and crafting materials no longer get a "KEEP" tag — there's no decision to make (you bank them and spend later), so the haul now saves its keep/sell verdicts for the drops that actually warrant one (enhancement recipes and salvage). Incarnate materials are still counted in your loot summary.
- **The Play Log now knows who you're playing — and links to their fit.** It reads the character name from the log ("playing Rattle") instead of only showing the account, tracks it across character switches, and attributes your stats to the right character. If that character has a saved build in Hero Companion, a one-click **"load their fit"** link appears — so switching from Rattle to Lime Juice brings up Lime Juice's build, not whoever you had open. (The game only names your character on a fresh login, so if it says "not detected yet," log out to character select and back in.) Because players sometimes rename characters, the fit link is one you can set explicitly — "link the open fit to this character" — so it sticks through a rename and fixes any wrong name-guess. No fit for a character yet? It offers to import their build straight from the game.

- **The Play Log's account switch is now obvious.** Which account you're watching shows as a row of chips (the active one highlighted with), and switching is one click on another chip — no more hunting for a "change account" link. For dual-boxers this matters: each account is its own log, so you flip between your two characters' stats instantly, and a note says so.
- **Your farm haul now knows your build.** When the Play Log is watching a character that has a saved fit, any drop whose set your build actually uses gets starred and floated to the top of the haul as a **"keep — for your build,"** naming the power that wants it. A standard set you would normally vendor becomes a keep when it is in your plan. Drops that do not fit still get the usual keep-or-sell advice.
- **Every power now explains how it is slotted.** Each power in a generated or solved build wears a small tag saying *why* it is slotted the way it is — a full set for its bonuses, a proc bomb (more AoE damage than a set here), global mules (uniques that each work from one slot), or a deliberate frankenslot. Hover it for the full reasoning. High-end slotting can look like scatter until you know the plan; now the build tells you the plan.
- **Fixed some warnings that were plain wrong.** The off-role check used to tell a Brute or Mastermind its powers "do not extend to damage dealer," which is nonsense — those are damage dealers and are now recognized as such. The "no damage enhancement" warning no longer fires on a power that is carrying a real set or damage procs. And Physical Perfection (and the epic-pool tier rules around it) now require the correct number of prerequisite powers, verified against a large corpus of real builds.
- **Better proc choices.** When the optimizer proc-bombs a power, it now ranks the available procs by actual expected damage, so the strongest procs land instead of whichever the data happened to list first.
- **The tray icon behaves.** Updating no longer leaves a "ghost" tray icon that only clears when you mouse over it — the old copy now shuts down cleanly and removes its own icon before the new one takes over. The tray menu also gains **Check for updates...** (it tells you right there whether a new version is out) alongside Open and Quit. Hero Companion keeps running in the tray after you close the browser tab so reopening is instant; the tray menu is how you drive or quit it.

0.12.7 — July 5, 2026

- **The Play Log parser is now built on real logs.** The first cut guessed at line formats; this version is validated against tens of thousands of lines of actual Homecoming chat logs. It reads XP and influence, enemies defeated (a real kill count), every drop sorted into recipes / salvage / incarnate materials / crafting mats with keep-or-sell advice, and Consignment House sales and purchases — while correctly ignoring the wall of combat, heal, and chat spam around them. Recipe drops map to their enhancement set for a proper verdict. (Veteran levels aren't written to the chat log by the game, so post-50 progress is shown as the incarnate materials you earn instead — that's what the log actually records.)

- **The Play Log reads live and walks you through setup.** Once you pick an account

it now updates on its own as you play (a "live" marker shows it's watching), so you don't have to keep clicking refresh. A built-in guide spells out the one-time in-game setup — make a chat tab named "Companion", add all channels to it, and turn on `/logchat` — so the only thing you have to remember is turning the log on. If there's no chat log for the day, the app says so and tells you how to start it. The opening screen is now framed simply: turn your sessions into *your* stats, all private and local — with any future community sharing kept as a separate choice. A plain privacy note spells out the promise: the tool only ever handles game data (prices, drops, stats), never your real name, email, location, IP, or account login, and never anything about the other players in your chat log.

- **The Play Log now asks before touching anything.** It read game files without

asking and without saying what happens to the data — against this app's own rules. Now the section opens with a plain statement: what it reads (the chat logs your game writes), what you get (insights), and the promise — everything stays on this computer, nothing is uploaded or shared, and any future optional sharing will be a separate, clearly labeled, anonymous opt-in. Until you click "Enable the Play Log", the app doesn't even look at the game's folders. "No thanks" is remembered and reversible.

- **Fixed: the Play Log could dead-end.** Clicking "Read new log entries" before

choosing an account replaced the account choices with an error message that told you to pick an account — with nothing left to pick. Now the read button simply doesn't exist until you're watching an account (it appears the moment you pick one), errors always bring the choices back, and the account choices look like buttons ("Watch filofinfain") instead of plain text.

0.12.6 — July 5, 2026

- **The Play Log (first cut).** A new full-width section at the bottom of the app

reads your in-game chat logs (type `/logchat` in game once to start writing them) and turns them into insight — no raw log ever shown. Pick which account to watch (multi-account players choose their main), hit "Read new log entries" after a session, and get: session totals (XP, influence in/out, merits, defeats), progress (level-ups, badges), and your recent haul with keep-vs-convert/sell verdicts on every drop. Reading is incremental — only new lines since last time. The line formats are provisional until validated against real logs, and the tool says so: it reports how much it parsed and shows any data-looking lines it didn't recognize, so early users help perfect it.

0.12.5 — July 5, 2026

- **Power trays now follow the community standard** (researched from player forums):

Tray 1 is your active rotation — openers, then the single-target chain in slots 1-3 (the muscle-memory keys), then AoEs and cones, with the nuke pinned at the end. Tray 2 is mid-fight clicks — self-buffs, then heals, then endurance recovery grouped together at the end. Tray 3 is set-and-forget — toggles in switch-on order (your armors first, then pools), utility, and Rest parked last. Tray 4 is travel and sprints, deliberately away from the combat keys. The same kind of power lands in the same place on every character, so your muscle memory transfers between alts — the way experienced players actually set up their bars.

- **The Incarnates brick.** Your six incarnate choices (the solver's recommendations, or your own picks) now live in a card of their own — same shape as the in-game power trays and docked right above them, with the real in-game incarnate icons (family art + rarity ring) and the full ability name on hover. Slots you haven't filled show as dashed placeholders.
- **Every IO icon now shows its level.** Set pieces the optimizer slots (and ones you pick by hand) carry the level to buy or craft them at — the set's maximum, so a Cloud Senses piece reads 30, an Achilles' Heel proc 20, premium sets 50 — right under the icon, the same way common IOs already did.
- **Procs are named properly.** Slots the proc pass fills used to read just "Annihilation: proc" — they now carry the real piece name ("Annihilation: Chance for Res Debuff"), so you know exactly what to buy.
- **Ctrl+Z undoes your last edit.** Mis-clicked a slot? One keystroke takes it back — same as the Undo button (power picks, slot changes, enhancement swaps, pool changes). Typing in a text box keeps its normal Ctrl+Z.

0.12.4 — July 4, 2026

- ****Level 1 always satisfies the creation requirement: one primary power AND one secondary power****, each from that set's first two — the basics come before anything else, on every archetype (same treatment as the epic-pool prerequisite rule). A Poison/Sonic Defender's level 1 is Shriek/Scream + Alkaloid/Envenom; Envenom-and-Alkaloid-both-at-1 can't happen anymore, anywhere it's displayed.
- **Old saves fix themselves.** A character saved by an earlier version could carry an impossible level-1 arrangement that survived every update because it was stored with the build. The app now re-checks the pick order every time it recalculates and quietly corrects it — open the build and it's right, no re-solve needed.

0.12.3 — July 4, 2026

- **Only one copy of the app can run.** Launching Hero Companion while it's already running now just opens the running copy's page instead of silently starting a second (or third) hidden copy on another port — the cause of "the app shows no difference after an update" reports: the browser was talking to a leftover old copy the whole time. After a self-update, any leftover old copies are shut down so the new version is the one you see.
- ****Fixed: a generated build's pick levels were dropped when loading an AI-designed tier****, so the build grid fell back to naive level badges (Alkaloid AND Envenom both shown at level 1). The seated pick levels now survive the load.

0.12.2 — July 4, 2026

- **Fixed: slots the game could never grant.** An enhancement slot is earned at a

specific level and can only be placed in a power you already have — so a power picked at level 49 can hold at most 4 slots (its free one plus the 3 earned at 50), and the 47 + 49 picks share just 6. The optimizer now seats heavily-slotted powers earlier in the pick order, re-solves with tighter budgets when even reordering can't fit (dense melee epics), and the validator calls out any hand-made arrangement the game would refuse ("picked at level 49 — it can hold at most 4 slots").

- **Smoother one-click updates.** The tab you click "Update now" in simply becomes the new version when the install finishes — the app no longer opens a duplicate tab while the old one lingers on the old version. (If that tab was closed, a fresh one still opens after a short wait.)
- **Level 1 is now framed as character creation.** In game your level-1 powers are one of the first two powers of your primary AND one of the first two of your secondary (verified against 2,255 master builds) — the plan now always seats a legal creation pair at level 1, the leveling walk says which choice was made and what the other option was, and the validator flags a build that skipped both of a set's first two powers.

0.12.1 — July 4, 2026

- **The decorative frame corners are gone.** The hero/villain identity now lives entirely in the color themes; the app uses the full window edge to edge.
- **Fixed: epic-pool pets taken without their prerequisites.** The game requires two other powers from an epic pool before its top-tier powers (Ice Elemental, Summon Spiderlings...) and one before the mid tier — the optimizer now takes the lower powers first, the search respects the ladder, and the validator counts precisely ("requires 2 other Ice Mastery powers — this build has 1").
- **Fixed: inherent powers shown as picks.** Health, Stamina, and other inherents are granted automatically by the game — the respec order no longer tells you to "take" them at a level (they get their own "automatic" footer line with slot counts), and their cards wear an "auto" badge instead of a level. Real picks' derived levels no longer shift to make room for them.

0.12.0 — July 4, 2026

The layout release — a complete redesign around one idea: a tight, horizontal, Sidekick-shaped workspace where every block fits snugly and every gap holds useful information.

- **The powers wall:** your build is a grid of uniform icon cards — power icon, name, level badge, enhancement icons with IO levels — flowing in pick order. No columns, no holes.
- **The build summary course:** beneath the wall, three equal bricks — Build Vitals (a labeled DEF/RES table plus recharge/recovery/HP/DPS), active Set Bonuses with stack counts and rule-of-five warnings, and the Uniques Carried checklist.
- **One tight masthead:** identity and icon-only actions in a single 40px strip.
- **Power Info panel:** click any power's name and a detail column opens on the right — type, availability, endurance cost, cast time, live damage/DPA/cycled-DPS (proc contributions included), allowed enhancement categories, and what's slotted in it right now. Numbers update with

every change.

- **Horizontal-first layout:** character setup (archetype, sets, pools, epic, incarnates) is a compact left rail with Stats and the Build Assistant stacked beneath it; the powers icon grid owns the entire wide side. The masthead is one tight strip — identity, vitals, icon-only actions — that stays pinned as you scroll.

- **No AI anywhere.** The panel is simply the Build Assistant — deterministic presets, goals, and instant Solve. (The AI seam still exists for those who bring their own key, but it's opt-in and invisible otherwise.)

- **The build is now an icon grid** (Sidekick-style): taken powers appear as compact cards — power icon, name, level badge — with their slotted enhancements as an icon row (IO level under each), arranged in three level columns (1–12 / 14–28 / 30–49). Add-power choices moved below the build. Same clicks as before: click a slot to change it, right-click to clear.

- **Overview bar:** your build's vitals — typed and positional defense, resistance, recharge, HP, ST/AoE DPS — in one color-coded line that stays visible while you scroll. Green means you've hit the current-meta mark.

- **Tighter layout:** on wide screens the stats and controls panels sit side by side (three columns total), paddings and text are denser — much more of the build fits in one view.

- During a one-click update, the old browser tab now says plainly that the app reopens in a new tab and this one can be closed — and it refreshes itself into the new version the moment you look at it (browsers throttle background tabs, which is why it previously seemed stuck on the old version).

0.11.1 — July 3, 2026

- **The optimizer now completes winter sets to 6 pieces** when typed defense is the goal — the big fire/cold defense bonuses the master builder flagged as missed are now chased properly (typed-defense targets carry full priority in the solver).

- **Wizard-built kits include procs by default.** A freshly generated build has no slotting to "preserve", so the proc pass now always runs on it.

0.11.0 — July 3, 2026

The master-class release — the optimizer went to school on 4,000+ builds shared by one of the community's best builders, plus his direct coaching.

- **Damage procs are now priced.** The engine computes each slotted %Damage proc's real contribution (PPM math: recharge, cast time, area factor), so the optimizer can genuinely trade set bonuses against procs — the core trade of the current meta.

- **Current-meta defense targets.** Presets now aim for ~35% Smashing/Lethal/Fire/Cold (typed) defense — or 35% Melee/Ranged/AoE for positional-armor characters — with the freed slots buying damage. Want the old 45% style? Put "classic softcap" in your goal. (Fire farm keeps its hard 45% fire floor — farms are still farms.)

- **Hasten gets 2 slots, never 3** — the third recharge IO is worth ~13% of face value

after diminishing returns. A wasted slot, now unwasted.

- **One-click updates** — "Update now" downloads and installs the new version for you; the app closes, updates, and restarts itself. No more trips to the download page.
- Model v24; scoring is stamped, so older champion scores aren't compared to new ones.

0.10.1 — July 3, 2026

- **Fixed: power tray icons missing in the installed app.** Three data files (power icons, the proc catalog, incarnate magnitudes) were loaded with a path that only worked when running from source — in the packaged app they silently loaded empty. The tray now shows real power icons, and the solver's proc pass has its full catalog back.
- **The app finds your in-game saves for you.** After `/build\_save\_file` in game, click "Find my characters for me" — Hero Companion scans the usual Homecoming install locations, lists every character save it finds (newest first), and imports on click. Unusual install? Paste your game folder once and it's remembered. The manual file picker remains as a fallback.

0.10.0 — July 3, 2026

The installer release — Hero Companion now behaves like a proper Windows app.

- **Real installer:** download `HeroCompanion-Setup-0.10.0.exe`, run it, done. It installs to a standard location, offers a desktop icon, adds a Start Menu entry, registers in Add/Remove Programs with a working uninstaller, and upgrades in place. (The portable zip still exists for folder-preferring folks.)
- **No more console window:** the app runs in the background with a tray icon (the lime pulse, next to your clock). Right-click it to open the app or quit. Closing the browser tab no longer strands a mystery window.
- **Updates button in the header** — check for updates on demand, whatever you answered at startup.
- Your saves live in %APPDATA%\HeroCompanion as always — installing, upgrading, or uninstalling never touches your characters.

0.9.1 — July 3, 2026

- **Update prompts (opt-in):** on first run the app asks once whether to check for updates at startup. Say yes and new releases show an "Update now / Remind me later" banner; say no and only the manual footer button ever checks.
- **Standalone builds are truly AI-free:** no more "AI: checking..." chip or "Ask Claude" box that could never work without a paid key. The panel is now the Build Assistant — presets, goals, and instant Solve, all fully offline. (Advanced: set HC_AI=1 with your own Claude key to re-enable the assistant.)
- **The app has its own icon** — the lime pulse badge, in the taskbar where it belongs, instead of the generic packaging icon.

- **Fixed:** the build wizard could clip its lower content on short windows with no way to scroll; it now scrolls within the dialog.
- **Fixed:** launching a second copy (or anything else holding port 5000) no longer kills the app — it moves to the next free port and says so.

0.9.0 — July 3, 2026

The "Hero Companion" release — the app has a name, a face, its papers in order, and a PC body to live in.

- **Standalone PC version:** Hero Companion now packages as a Windows executable — no Python, no setup; run it and the app opens in your browser. Your saves live in %APPDATA%\HeroCompanion.
- **Report a bug:** one click opens a pre-filled report (versions auto-included) on the project's GitHub page. Nothing is ever sent without your click.
- **Submit champion:** beat a shipped champion build? Export your candidate and post it to the submission queue — verified wins become the next shipped champion, with credit.
- **Check for updates:** compares your version against the latest GitHub release. Never automatic, never forced.
- **Hero/Villain reskin:** the app is now *Hero Companion* (or *Villain Companion* — flip alignment from the banner). Themed corners, colors, and cards; your choice is remembered.
- **Soldiers & Widows leveled honestly:** the level-1-to-50 walk now follows the real VEAT career — base powers only until the mandatory level-24 respect, then the full re-place order with your Crab/Bane or Night Widow/Fortunata branch.
- **Kheldians leveled honestly:** single-path sets auto-select, inherent flight means "no extra travel" by default, and the epic-pool step is gone (they don't get one).
- **Enhancement Converter planner + haul appraiser:** paste your drops (recipes included) and get keep/convert/sell verdicts with concrete converter paths and merit costs.
- **Origin pool rule enforced:** only one of Sorcery / Experimentation / Force of Will / Gadgetry / Utility Belt per build — the optimizer and validator both know the game won't allow a second.
- **Optimizer explores all pools:** Sorcery, Experimentation, and friends are now real candidates in the search, not just Fighting/Leadership/Speed/Leaping.
- **Terms, License, Credits:** CC BY-NC-SA 4.0 (free & noncommercial forever), Terms of Use, and full credits — including Guyver [SoV] of Sovereign, whose master builds were this tool's calibration standard.
- **This help system:** the Help button and this document.

Earlier (the road here)

- Physics/encounter model v23: wiki-verified to-hit, resistance, purple patch, mez protection, and archvillain rules; role-weighted scoring; scenario coverage.

- Deep optimizer: full-neighborhood search that runs to convergence with honest certificates, warm-started from champion builds, learning across runs.
- Leveling companion: per-level walk of the exact Homecoming schedule with cost-smart enhancement advice, deviation tracking, and end-game re-fit.
- Mids Reborn import/export compatibility.